Estimating the Number of Nodes and Paths in a DAG by Following a Random Path—Extensive Tables of Experimental Results

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											Permutations	
	$n=7$			$n=8$			$n=9$			$n=10$		
m	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$
10^1	$4{,}122.9 \pm 9{,}304.0$	3.03×10^4	0.73462	$39{,}331.8 \pm 59{,}061.9$	1.42×10^5	0.36044	$48{,}168.7 \pm 56{,}348.3$	1.68×10^5	0.34803	$41{,}437.8 \pm 37{,}261.1$	1.18×10^5	0.28431
10^2	$3{,}444.5 \pm 6{,}008.7$	3.35×10^4	0.09723	$28{,}772.5 \pm 64{,}855.0$	3.42×10^5	0.11875	$134{,}918.1 \pm 4.87\times10^5$	4.21×10^6	0.31195	$404{,}610.8 \pm 1.29\times10^6$	9.64×10^6	0.23824
10^3	$4{,}558.2 \pm 14{,}350.9$	1.95×10^5	0.04285	$77{,}136.0 \pm 1.55\times10^6$	4.83×10^7	0.62640	$188{,}542.8 \pm 1.33\times10^6$	2.93×10^7	0.15514	$7{,}137{,}705.1 \pm 1.99\times10^8$	6.29×10^9	0.88158
10^4	$5{,}352.3 \pm 27{,}948.9$	1.29×10^6	0.02416	$44{,}629.3 \pm 6.57\times10^5$	4.83×10^7	0.10826	$221{,}679.5 \pm 2.23\times10^6$	1.49×10^8	0.06720	$2{,}335{,}438.1 \pm 6.86\times10^7$	6.29×10^9	0.26943
10^5	$5{,}131.4 \pm 29{,}457.4$	3.51×10^6	0.00685	$42{,}382.8 \pm 1.13\times10^6$	2.89×10^8	0.06821	$372{,}754.8 \pm 1.75\times10^7$	4.94×10^9	0.13250	$2{,}920{,}867.5 \pm 1.47\times10^8$	2.87×10^{10}	0.09840
10^6	$5{,}016.9 \pm 40{,}230.5$	2.23×10^7	0.00445	$39{,}837.5 \pm 8.32\times10^5$	3.63×10^8	0.00912	$347{,}645.5 \pm 1.23\times10^7$	4.98×10^9	0.01433	$3{,}016{,}115.1 \pm 2.69\times10^8$	1.73×10^{11}	0.05721
10^7	$5{,}046.5 \pm 38{,}894.2$	3.39×10^7	0.00067	$40{,}820.6 \pm 1.92\times10^6$	5.16×10^9	0.01265	$351{,}350.7 \pm 2.59\times10^7$	4.64×10^{10}	0.01320	$3{,}281{,}363.3 \pm 6.06\times10^8$	1.34×10^{12}	0.04080
	$5{,}040 \pm 3.730\times10^4$	#visited = 22		$40{,}320 \pm 1.642\times10^6$	#visited = 29		$362{,}880 \pm 1.614\times10^8$	#visited = 37		$3{,}628{,}800 \pm 3.682\times10^{10}$	#visited = 46	
	$39{,}916{,}800 \pm 2.010\times10^{13}$	#visited = 56		$479{,}001{,}600 \pm 2.720\times10^{16}$	#visited = 67							

Table 1: Algorithm P, Path Sampling, permutations

Permutations												
	$n=7$			$n=8$			$n=9$			$n=10$		
m	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$
10^1	$2{,}134.2 \pm 2{,}895.3$	1.00×10^4	0.46900	$10{,}564.8 \pm 11{,}944.9$	3.32×10^4	0.31394	$234{,}362.6 \pm 4.41\times10^5$	1.28×10^6	0.54434	$485{,}847.2 \pm 1.01\times10^6$	3.33×10^6	0.68531
10^2	$5{,}735.4 \pm 14{,}589.7$	1.30×10^5	0.22703	$15{,}677.9 \pm 28{,}749.5$	1.85×10^5	0.11808	$315{,}348.3 \pm 1.03\times10^6$	6.34×10^6	0.20111	$1{,}513{,}525.6 \pm 6.94\times10^6$	5.11×10^7	0.33763
10^3	$5{,}322.5 \pm 17{,}623.9$	3.56×10^5	0.06698	$47{,}912.4 \pm 4.33\times10^5$	1.23×10^7	0.25653	$281{,}586.0 \pm 1.10\times10^6$	1.40×10^7	0.04980	$2{,}192{,}325.9 \pm 2.70\times10^7$	8.20×10^8	0.37425
10^4	$5{,}198.4 \pm 21{,}658.4$	1.21×10^6	0.02331	$39{,}265.0 \pm 2.24\times10^5$	1.23×10^7	0.03130	$312{,}165.1 \pm 3.48\times10^6$	2.36×10^8	0.07569	$5{,}269{,}581.7 \pm 2.24\times10^8$	1.82×10^{10}	0.34561
10^5	$4{,}949.1 \pm 17{,}630.6$	2.18×10^6	0.00440	$40{,}970.8 \pm 3.82\times10^5$	6.63×10^7	0.01619	$411{,}450.9 \pm 2.05\times10^7$	6.05×10^9	0.14698	$3{,}143{,}399.0 \pm 1.01\times10^8$	1.82×10^{10}	0.05794
10^6	$5{,}010.6 \pm 17{,}626.0$	3.99×10^6	0.00080	$39{,}777.6 \pm 4.31\times10^5$	2.35×10^8	0.00592	$360{,}495.0 \pm 1.13\times10^7$	6.05×10^9	0.01678	$3{,}225{,}251.6 \pm 1.37\times10^8$	5.67×10^{10}	0.01757
10^7	$5{,}033.0 \pm 18{,}375.8$	5.41×10^6	0.00011	$40{,}289.8 \pm 5.41\times10^5$	7.25×10^8	0.00180	$372{,}028.6 \pm 4.02\times10^7$	1.01×10^{11}	0.02708	$3{,}335{,}166.5 \pm 2.18\times10^8$	2.83×10^{11}	0.00848
	$5{,}040 \pm 1.883\times10^4$	#visited = 22		$40{,}320 \pm 6.316\times10^5$	#visited = 29		$362{,}880 \pm 4.665\times10^7$	#visited = 37		$3{,}628{,}800 \pm 8.110\times10^9$	#visited = 46	
	$39{,}916{,}800 \pm 3.400\times10^{12}$	#visited = 56		$479{,}001{,}600 \pm 3.540\times10^{15}$	#visited = 67							

Table 2: IS by outdegree, permutations

	Permutations											
	$n = 7$			$n = 8$			$n = 9$			$n = 10$		
m	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.		
10^1	3,401.0 ± 2,381.2	9.00×10^3	0.26455	41,711.3 ± 68,536.3	2.12×10^5	0.50920	$120{,}931.2 \pm 1.24\times10^5$	4.52×10^5	0.37377	$1{,}304{,}852.6 \pm 1.58\times10^6$		4
10^2	4,972.0 ± 9,786.2	6.72×10^4	0.13508	$39{,}047.5 \pm 1.40\times10^5$	1.36×10^6	0.34866	$258{,}487.9 \pm 9.36\times10^5$	8.60×10^6	0.33284	$1{,}989{,}932.1 \pm 6.84\times10^6$		6
10^3	5,297.5 ± 10,317.7	1.53×10^5	0.02888	34,062.1 ± 89,284.0	1.73×10^6	0.05073	$398{,}791.7 \pm 3.63\times10^6$	1.08×10^8	0.27097	$5{,}555{,}099.3 \pm 9.55\times10^7$		3
10^4	4,982.0 ± 9,559.2	2.39×10^5	0.00480	$38{,}986.7 \pm 1.34\times10^5$	5.97×10^6	0.01532	$437{,}469.8 \pm 1.12\times10^7$	1.09×10^9	0.25028	$3{,}176{,}269.7 \pm 4.27\times10^7$		3
10^5	5,029.7 ± 10,465.5	6.12×10^5	0.00122	$39{,}853.5 \pm 2.02\times10^5$	4.61×10^7	0.01157	$357{,}426.5 \pm 4.90\times10^6$	1.09×10^9	0.03063	$3{,}117{,}811.0 \pm 3.77\times10^7$		5
10^6	5,039.7 ± 10,442.2	1.26×10^6	0.00025	$40{,}025.5 \pm 2.55\times10^5$	1.28×10^8	0.00321	$360{,}984.5 \pm 5.31\times10^6$	2.67×10^9	0.00740	$3{,}550{,}915.3 \pm 2.45\times10^8$		2
	5,040 #visited: avg = 43.7, max = 117 #generated: avg = 132.6			40,320 #visited: avg = 84.4, max = 277 #generated: avg = 290.6			362,880 #visited: avg = 180.7, max = 763 #generated: avg = 690.5			3,628,800 #visited: avg = 432.1, max #generated: avg = 1822.1		

	Permutations											
			$n = 10$			$n = 11$			$n = 12$			
ax.	$\hat{\chi}$		mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	
$\times10^5$	0.37377		$1{,}304{,}852.6 \pm 1.58\times10^6$	4.31×10^6	0.33045	$15{,}200{,}342.5 \pm 2.95\times10^7$	8.17×10^7	0.53724	$37{,}282{,}864.7 \pm 4.75\times10^7$	1.30×10^8	0.34881	
$\times10^6$	0.33284		$1{,}989{,}932.1 \pm 6.84\times10^6$	6.47×10^7	0.32493	$14{,}272{,}656.7 \pm 4.50\times10^7$	3.83×10^8	0.26829	$371{,}709{,}343.5 \pm 1.77\times10^9$	1.35×10^{10}	0.36311	
$\times10^8$	0.27097		$5{,}555{,}099.3 \pm 9.55\times10^7$	3.00×10^9	0.54003	$32{,}993{,}920.2 \pm 3.25\times10^8$	7.36×10^9	0.22311	$213{,}058{,}237.5 \pm 2.44\times10^9$	7.20×10^{10}	0.33784	
$\times10^9$	0.25028		$3{,}176{,}269.7 \pm 4.27\times10^7$	3.00×10^9	0.09445	$26{,}030{,}745.1 \pm 2.83\times10^8$	1.52×10^{10}	0.05850	$288{,}364{,}656.4 \pm 7.78\times10^9$	7.04×10^{11}	0.24413	
$\times10^9$	0.03063		$3{,}117{,}811.0 \pm 3.77\times10^7$	5.11×10^9	0.01637	$30{,}516{,}913.9 \pm 5.87\times10^8$	1.07×10^{11}	0.03505	$396{,}763{,}404.5 \pm 2.76\times10^{10}$	7.72×10^{12}	0.19467	
$\times10^9$	0.00740		$3{,}550{,}915.3 \pm 2.45\times10^8$	2.38×10^{11}	0.06694	$39{,}677{,}524.5 \pm 2.67\times10^9$	1.76×10^{12}	0.04446	$356{,}575{,}566.3 \pm 2.51\times10^{10}$	1.31×10^{13}	0.03679	
= 763			3,628,800 #visited: avg = 432.1, max = 2252 #generated: avg = 1822.1			39,916,800 #visited: avg = 1184.0, max = 7807 #generated: avg = 5521.6			479,001,600 #visited: avg = 3723.2, max = 30890 #generated: avg = 19255.0			

Table 3: Bucket Sampling, permutations

	Permutations											
	$n = 7$			$n = 8$			$n = 9$			$n = 10$		
m	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.		
10^1	4,402.8 ± 1,685.3	8.69×10^3	0.19734	51,102.7 ± 43,393.4	1.56×10^5	0.30491	199,951.9 ± 1.45×10^5	4.06×10^5	0.20327	992,141.6 ± 8.10×10^5		2.6
10^2	4,805.4 ± 2,436.1	1.73×10^4	0.03610	43,250.2 ± 33,729.6	1.56×10^5	0.03603	370,850.2 ± 3.97×10^5	2.05×10^6	0.05538	3,128,298.4 ± 5.38×10^6		3.4
10^3	5,018.2 ± 2,366.6	1.85×10^4	0.00369	43,252.0 ± 39,430.2	3.70×10^5	0.00855	317,016.9 ± 3.77×10^5	3.08×10^6	0.00971	3,392,416.3 ± 7.61×10^6		9.2
10^4	5,072.7 ± 2,506.7	2.76×10^4	0.00054	40,068.4 ± 35,080.8	7.40×10^5	0.00185	359,232.3 ± 5.80×10^5	1.84×10^7	0.00513	3,650,094.3 ± 1.24×10^7		7.4
10^5	5,043.2 ± 2,459.5	3.64×10^4	0.00007	40,222.5 ± 35,171.4	8.77×10^5	0.00022	362,323.6 ± 5.76×10^5	2.37×10^7	0.00066	3,610,817.9 ± 1.01×10^7		7.8
10^6	5,042.1 ± 2,447.1	3.98×10^4	0.00001	40,259.1 ± 35,177.0	1.37×10^6	0.00003	361,797.9 ± 5.81×10^5	7.45×10^7	0.00021	3,629,968.8 ± 1.18×10^7		2.5
	5,040 #visited: avg = 190.5, max = 194 #generated: avg = 560.5 #saved: avg = 125.5 = 22.4%			40,320 #visited: avg = 261.1, max = 266 #generated: avg = 893.4 #saved: avg = 178.7 = 20.0%			362,880 #visited: avg = 341.8, max = 348 #generated: avg = 1329.4 #saved: avg = 240.4 = 18.1%			3,628,800 #visited: avg = 432.7, max #generated: avg = 1879.1 #saved: avg = 308.9 = 16.4		

Permutations											
		$n = 10$			$n = 11$			$n = 12$			
max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	
6×10^5	0.20327	992,141.6 ± 8.10×10^5	2.66×10^6	0.26765	28,457,101.8 ± 6.14×10^7	2.02×10^8	0.71002	94,776,847.4 ± 1.34×10^8	4.58×10^8	0.48371	
5×10^6	0.05538	3,128,298.4 ± 5.38×10^6	3.41×10^7	0.10893	39,945,262.1 ± 9.95×10^7	6.74×10^8	0.16869	657,082,732.9 ± 4.05×10^9	4.05×10^{10}	0.61589	
8×10^6	0.00971	3,392,416.3 ± 7.61×10^6	9.27×10^7	0.02732	33,867,322.4 ± 1.59×10^8	4.40×10^9	0.12978	318,226,359.9 ± 1.50×10^9	4.05×10^{10}	0.12717	
4×10^7	0.00513	3,650,094.3 ± 1.24×10^7	7.46×10^8	0.02043	37,623,804.0 ± 1.76×10^8	1.03×10^{10}	0.02727	532,263,222.0 ± 7.71×10^9	4.76×10^{11}	0.08934	
7×10^7	0.00066	3,610,817.9 ± 1.01×10^7	7.87×10^8	0.00218	39,645,227.3 ± 2.89×10^8	4.22×10^{10}	0.01064	474,798,274.8 ± 9.91×10^9	2.76×10^{12}	0.05808	
5×10^7	0.00021	3,629,968.8 ± 1.18×10^7	2.56×10^9	0.00070	39,879,356.1 ± 3.08×10^8	1.33×10^{11}	0.00334	469,392,213.3 ± 7.36×10^9	2.76×10^{12}	0.00587	
x = 348 .1%		3,628,800 #visited: avg = 432.7, max = 440 #generated: avg = 1879.1 #saved: avg = 308.9 = 16.4%			39,916,800 #visited: avg = 533.7, max = 541 #generated: avg = 2553.0 #saved: avg = 383.3 = 15.0%			479,001,600 #visited: avg = 643.7, max = 650 #generated: avg = 3350.2 #saved: avg = 455.5 = 13.6%			

Table 4: Crowd Sampling, permutations, budget $B = 10$

Klee–Minty cubes												
	$n = 8$			$n = 10$			$n = 12$			$n = 14$		
m	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$
10^1	52.9 ± 57.3	1.89×10^2	0.35831	$795.5 \pm 2{,}170.0$	6.96×10^3	0.87488	198.8 ± 335.8	1.05×10^3	0.53028	157.2 ± 239.4	7.82×10^2	0.49757
10^2	136.9 ± 658.3	6.29×10^3	0.45975	277.6 ± 937.2	6.96×10^3	0.25073	$771.3 \pm 3{,}045.2$	2.41×10^4	0.31309	$1{,}116.5 \pm 4{,}644.2$	3.79×10^4	0.33948
10^3	147.9 ± 953.5	2.63×10^4	0.17788	$906.3 \pm 11{,}238.4$	2.85×10^5	0.31473	$1{,}068.5 \pm 10{,}497.3$	2.81×10^5	0.26334	$1{,}061.3 \pm 8{,}224.7$	2.26×10^5	0.21306
10^4	$189.2 \pm 2{,}317.7$	1.89×10^5	0.10008	$485.6 \pm 4{,}998.2$	2.85×10^5	0.05874	$1{,}902.3 \pm 85{,}597.0$	8.39×10^6	0.44087	$2{,}851.2 \pm 1.11\times10^5$	1.09×10^7	0.38278
10^5	$224.2 \pm 3{,}882.0$	6.31×10^5	0.02817	$955.3 \pm 90{,}694.9$	2.81×10^7	0.29453	$1{,}786.1 \pm 60{,}458.6$	9.70×10^6	0.05430	$8{,}379.5 \pm 1.38\times10^6$	4.33×10^8	0.51729
10^6	$286.5 \pm 35{,}111.9$	2.70×10^7	0.09438	$795.1 \pm 59{,}515.4$	4.42×10^7	0.05565	$2{,}206.7 \pm 2.13\times10^5$	1.52×10^8	0.06900	$6{,}436.0 \pm 8.52\times10^5$	4.33×10^8	0.06735
10^7	$290.3 \pm 1.11\times10^5$	3.43×10^8	0.11804	$909.0 \pm 1.78\times10^5$	4.89×10^8	0.05384	$3{,}103.9 \pm 2.13\times10^6$	6.58×10^9	0.21208	$9{,}387.8 \pm 7.90\times10^6$	2.45×10^{10}	0.26058
	256 $\pm 1.362\times10^5$			1,024 $\pm 3.246\times10^9$			4,096 $\pm 1.034\times10^{19}$			16,384 $\pm 2.024\times10^{39}$		
	#visited: avg = 11.5, max = 50			#visited: avg = 15.7, max = 78			#visited: avg = 20.6, max = 100			#visited: avg = 26.4, max = 124		

Klee–Minty cubes												
	$n = 12$			$n = 14$			$n = 16$			$n = 18$		
ev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$
	1.05×10^3	0.53028	157.2 ± 239.4	7.82×10^2	0.49757	$787.1 \pm 1{,}151.5$	3.44×10^3	0.43755	$41{,}739.1 \pm 1.21\times10^5$	3.84×10^5	0.92002
2	2.41×10^4	0.31309	$1{,}116.5 \pm 4{,}644.2$	3.79×10^4	0.33948	$956.0 \pm 3{,}225.2$	2.24×10^4	0.23416	$48{,}095.3 \pm 4.15\times10^5$	4.14×10^6	0.86007
7.3	2.81×10^5	0.26334	$1{,}061.3 \pm 8{,}224.7$	2.26×10^5	0.21306	$3{,}976.2 \pm 44{,}371.1$	1.05×10^6	0.26482	$11{,}616.2 \pm 1.75\times10^5$	4.14×10^6	0.35610
7.0	8.39×10^6	0.44087	$2{,}851.2 \pm 1.11\times10^5$	1.09×10^7	0.38278	$4{,}816.5 \pm 78{,}003.1$	4.57×10^6	0.09481	$19{,}783.3 \pm 1.04\times10^6$	1.03×10^8	0.51881
8.6	9.70×10^6	0.05430	$8{,}379.5 \pm 1.38\times10^6$	4.33×10^8	0.51729	$22{,}698.0 \pm 3.26\times10^6$	9.07×10^8	0.39944	$16{,}121.4 \pm 8.24\times10^5$	1.47×10^8	0.09139
10^5	1.52×10^8	0.06900	$6{,}436.0 \pm 8.52\times10^5$	4.33×10^8	0.06735	$10{,}246.3 \pm 1.19\times10^6$	9.07×10^8	0.08849	$53{,}023.0 \pm 2.93\times10^7$	2.91×10^{10}	0.54946
10^6	6.58×10^9	0.21208	$9{,}387.8 \pm 7.90\times10^6$	2.45×10^{10}	0.26058	$127{,}868.8 \pm 3.49\times10^8$	1.10×10^{12}	0.86258	$43{,}846.1 \pm 2.12\times10^7$	5.09×10^{10}	0.11607
$\times10^{19}$			16,384 $\pm 2.024\times10^{39}$			65,536 $\pm 3.055\times10^{83}$			262,144 $\pm$ (overflow)		
= 20.6, max = 100			#visited: avg = 26.4, max = 124			#visited: avg = 33.0, max = 152			#visited: avg = 40.5, max = 186		

Table 5: Algorithm P, Path Sampling, Klee-Minty cubes

Klee–Minty cubes												
	$n = 8$			$n = 10$			$n = 12$			$n = 14$		
m	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$
10^1	117.7 ± 115.1	3.22×10^2	0.27400	65.0 ± 46.7	1.45×10^2	0.22366	503.5 ± 668.7	1.99×10^3	0.39488	$860.4 \pm 1{,}517.7$	4.99×10^3	0.57991
10^2	196.5 ± 426.6	2.76×10^3	0.14069	$500.7 \pm 1{,}279.9$	7.67×10^3	0.15319	$685.5 \pm 1{,}676.9$	1.05×10^4	0.15256	$1{,}978.2 \pm 9{,}092.5$	8.94×10^4	0.45210
10^3	$225.2 \pm 1{,}547.5$	4.57×10^4	0.20286	$649.7 \pm 3{,}306.4$	5.54×10^4	0.08534	$1{,}854.5 \pm 15{,}942.5$	3.76×10^5	0.20282	$2{,}437.9 \pm 13{,}595.7$	2.87×10^5	0.11754
10^4	$221.7 \pm 1{,}030.9$	4.57×10^4	0.02060	$880.2 \pm 11{,}958.6$	8.46×10^5	0.09610	$2{,}418.5 \pm 33{,}458.9$	2.61×10^6	0.10777	$6{,}236.8 \pm 1.16\times10^5$	8.23×10^6	0.13192
10^5	$246.1 \pm 2{,}141.0$	3.77×10^5	0.01533	$910.0 \pm 37{,}608.7$	1.16×10^7	0.12699	$6{,}892.9 \pm 1.23\times10^6$	3.86×10^8	0.55991	$8{,}027.4 \pm 2.10\times10^5$	2.95×10^7	0.03677
10^6	$250.2 \pm 3{,}524.5$	1.68×10^6	0.00670	$983.9 \pm 87{,}220.8$	8.33×10^7	0.08466	$3{,}969.2 \pm 5.15\times10^5$	3.86×10^8	0.09723	$11{,}023.0 \pm 1.47\times10^6$	1.12×10^9	0.10146
10^7	$254.6 \pm 4{,}391.8$	7.87×10^6	0.00309	$1{,}006.4 \pm 1.45\times10^5$	4.38×10^8	0.04347	$3{,}737.7 \pm 4.70\times10^5$	7.03×10^8	0.01881	$10{,}627.0 \pm 1.42\times10^6$	3.21×10^9	0.03018
	256 $\pm 6.559\times10^3$			1,024 $\pm 2.141\times10^6$			4,096 $\pm 7.762\times10^{10}$			16,384 $\pm 5.948\times10^{19}$		
	#visited: avg = 16.3, max = 60			#visited: avg = 22.4, max = 86			#visited: avg = 29.2, max = 104			#visited: avg = 36.8, max = 13		

Klee–Minty cubes												
$n = 12$			$n = 14$			$n = 16$			$n = 18$			
. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	
3.7	1.99×10^3	0.39488	$860.4 \pm 1{,}517.7$	4.99×10^3	0.57991	690.7 ± 873.0	2.52×10^3	0.36420	$4{,}325.1 \pm 9{,}754.6$	3.17×10^4	0.73304	
76.9	1.05×10^4	0.15256	$1{,}978.2 \pm 9{,}092.5$	8.94×10^4	0.45210	$2{,}977.4 \pm 11{,}559.9$	9.74×10^4	0.32699	$8{,}147.2 \pm 41{,}240.2$	3.78×10^5	0.46370	
942.5	3.76×10^5	0.20282	$2{,}437.9 \pm 13{,}595.7$	2.87×10^5	0.11754	$36{,}048.5 \pm 6.29\times10^5$	1.68×10^7	0.46715	$16{,}867.4 \pm 2.37\times10^5$	6.86×10^6	0.40655	
458.9	2.61×10^6	0.10777	$6{,}236.8 \pm 1.16\times10^5$	8.23×10^6	0.13192	$14{,}215.6 \pm 2.66\times10^5$	1.68×10^7	0.11846	$23{,}893.3 \pm 3.70\times10^5$	1.74×10^7	0.07291	
3×10^6	3.86×10^8	0.55991	$8{,}027.4 \pm 2.10\times10^5$	2.95×10^7	0.03677	$21{,}463.1 \pm 9.90\times10^5$	2.21×10^8	0.10303	$35{,}524.0 \pm 1.19\times10^6$	2.69×10^8	0.07572	
5×10^5	3.86×10^8	0.09723	$11{,}023.0 \pm 1.47\times10^6$	1.12×10^9	0.10146	$28{,}434.1 \pm 2.70\times10^6$	1.30×10^9	0.04584	$90{,}993.0 \pm 2.34\times10^7$	1.91×10^{10}	0.21026	
0×10^5	7.03×10^8	0.01881	$10{,}627.0 \pm 1.42\times10^6$	3.21×10^9	0.03018	$31{,}275.0 \pm 4.92\times10^6$	8.05×10^9	0.02575	$84{,}114.1 \pm 2.28\times10^7$	4.30×10^{10}	0.05117	
62×10^{10}			16,384 $\pm 5.948\times10^{19}$			65,536 $\pm 6.361\times10^{36}$			262,144 $\pm 1.641\times10^{69}$			
g = 29.2, max = 104			#visited: avg = 36.8, max = 134			#visited: avg = 45.3, max = 160			#visited: avg = 54.6, max = 202			

Table 6: IS by outdegree, Klee-Minty cubes

Klee–Minty cubes											
	$n = 8$			$n = 10$			$n = 12$			$n = 14$	
m	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.
10^1	157.3 ± 176.2	5.21×10^2	0.33112	218.6 ± 226.5	7.47×10^2	0.34168	472.9 ± 661.9	2.03×10^3	0.43003	9,500.4 ± 27,304.3	8.69×10^4
10^2	290.8 ± 1,204.6	1.11×10^4	0.38257	640.6 ± 1,830.8	1.30×10^4	0.20285	2,626.8 ± 11,882.5	8.69×10^4	0.33074	9,645.5 ± 64,178.9	6.29×10^5
10^3	192.6 ± 663.0	1.21×10^4	0.06259	756.1 ± 5,094.5	1.33×10^5	0.17623	2,167.0 ± 27,495.6	8.44×10^5	0.38950	4,023.5 ± 30,694.3	6.29×10^5
10^4	238.5 ± 1,608.7	1.02×10^5	0.04293	818.4 ± 9,017.6	5.12×10^5	0.06258	2,403.3 ± 36,794.5	2.26×10^6	0.09389	4,751.1 ± 65,667.4	3.97×10^6
10^5	243.7 ± 1,977.1	3.86×10^5	0.01585	891.8 ± 20,241.5	5.13×10^6	0.05748	4,349.1 ± 5.99×10^5	1.88×10^8	0.43315	7,907.6 ± 3.47×10^5	7.14×10^7
10^6	253.4 ± 2,945.6	1.06×10^6	0.00418	984.7 ± 31,787.0	1.26×10^7	0.01279	6,346.5 ± 2.53×10^6	2.35×10^9	0.37095	10,052.8 ± 1.74×10^6	1.52×10^9
10^7	259.5 ± 8,700.2	2.30×10^7	0.00886	982.6 ± 50,113.4	6.09×10^7	0.00620	4,877.9 ± 3.17×10^6	9.30×10^9	0.19071	11,432.1 ± 2.86×10^6	6.22×10^9
	256 ± 5.045×10^3			1,024 ± 2.080×10^6			4,096 ± 6.025×10^{11}			16,384 ± 1.718×10^{24}	
	#visited: avg = 25.4, max = 72			#visited: avg = 37.8, max = 108			#visited: avg = 53.0, max = 148			#visited: avg = 71.0, max = 186	

Klee–Minty cubes											
$n = 12$			$n = 14$			$n = 16$			$n = 18$		
dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$
9	2.03×10^3	0.43003	9,500.4 ± 27,304.3	8.69×10^4	0.91501	524.2 ± 853.6	2.57×10^3	0.48954	1,454.6 ± 2,572.9	7.52×10^3	0.51694
32.5	8.69×10^4	0.33074	9,645.5 ± 64,178.9	6.29×10^5	0.65231	2,259.4 ± 6,631.5	4.56×10^4	0.20167	10,847.7 ± 78,752.8	7.86×10^5	0.72444
95.6	8.44×10^5	0.38950	4,023.5 ± 30,694.3	6.29×10^5	0.15638	8,649.3 ± 1.09×10^5	2.84×10^6	0.32811	238,611.2 ± 7.34×10^6	2.32×10^8	0.97223
94.5	2.26×10^6	0.09389	4,751.1 ± 65,667.4	3.97×10^6	0.08359	56,089.9 ± 3.16×10^6	2.84×10^8	0.50615	36,486.1 ± 2.33×10^6	2.32×10^8	0.63582
$\times10^5$	1.88×10^8	0.43315	7,907.6 ± 3.47×10^5	7.14×10^7	0.09027	17,128.0 ± 1.07×10^6	2.84×10^8	0.16575	40,350.6 ± 2.40×10^6	4.15×10^8	0.10288
$\times10^6$	2.35×10^9	0.37095	10,052.8 ± 1.74×10^6	1.52×10^9	0.15072	589,345.6 ± 5.66×10^8	5.66×10^{11}	0.96076	71,080.6 ± 1.92×10^7	1.65×10^{10}	0.23229
$\times10^6$	9.30×10^9	0.19071	11,432.1 ± 2.86×10^6	6.22×10^9	0.05441	86,284.4 ± 1.79×10^8	5.66×10^{11}	0.65622	104,257.7 ± 6.83×10^7	1.51×10^{11}	0.14454
5×10^{11}			16,384 ± 1.718×10^{24}			65,536 ± 3.318×10^{51}			262,144 ± 4.872×10^{113}		
= 53.0, max = 148			#visited: avg = 71.0, max = 186			#visited: avg = 92.0, max = 232			#visited: avg = 116.0, max = 292		

Table 7: IS by size, Klee-Minty cubes

Klee–Minty cubes											
	$n = 8$			$n = 10$			$n = 12$			$n = 14$	
m	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.
10^1	122.1 ± 60.4	2.46×10^2	0.20176	607.9 ± 396.9	1.28×10^3	0.21010	$868.0 \pm 1{,}456.5$	4.92×10^3	0.56699	$1{,}310.2 \pm 1{,}715.9$	6.10×10^3
10^2	235.6 ± 217.4	1.17×10^3	0.04964	$857.0 \pm 1{,}552.7$	1.27×10^4	0.14858	$1{,}704.5 \pm 4{,}347.4$	4.04×10^4	0.23701	$5{,}664.7 \pm 13{,}197.9$	9.47×10^4
10^3	239.5 ± 344.0	5.45×10^3	0.02275	$985.6 \pm 3{,}287.0$	5.65×10^4	0.05729	$3{,}150.8 \pm 13{,}846.9$	2.47×10^5	0.07848	$22{,}522.2 \pm 4.64\times10^5$	1.46×10^7
10^4	253.1 ± 500.0	2.09×10^4	0.00825	$971.5 \pm 3{,}667.1$	1.16×10^5	0.01198	$3{,}677.1 \pm 37{,}719.8$	3.33×10^6	0.09052	$28{,}023.4 \pm 1.57\times10^6$	1.56×10^8
10^5	256.3 ± 565.4	9.30×10^4	0.00363	$1{,}031.9 \pm 8{,}148.2$	1.93×10^6	0.01874	$3{,}717.0 \pm 52{,}576.6$	1.26×10^7	0.03399	$20{,}509.9 \pm 1.87\times10^6$	5.65×10^8
10^6	256.6 ± 541.4	9.30×10^4	0.00036	$1{,}029.4 \pm 11{,}760.9$	9.25×10^6	0.00899	$4{,}268.1 \pm 2.40\times10^5$	2.18×10^8	0.05108	$15{,}669.2 \pm 8.35\times10^5$	5.65×10^8
10^7	255.9 ± 533.7	1.05×10^5	0.00004	$1{,}025.8 \pm 9{,}644.3$	1.21×10^7	0.00117	$4{,}053.0 \pm 1.22\times10^5$	2.18×10^8	0.00538	$15{,}889.1 \pm 1.21\times10^6$	1.97×10^9
	256 $\pm 5.394\times10^2$			1,024 $\pm 1.071\times10^4$			4,096 $\pm 9.419\times10^5$			16,384 $\pm 1.692\times10^9$	
	#visited: avg = 28.9, max = 82			#visited: avg = 42.5, max = 112			#visited: avg = 58.7, max = 152			#visited: avg = 77.6, max = 202	

Klee–Minty cubes											
$n = 12$			$n = 14$			$n = 16$			$n = 18$		
dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$
6.5	4.92×10^3	0.56699	$1{,}310.2 \pm 1{,}715.9$	6.10×10^3	0.46544	$127{,}609.1 \pm 2.61\times10^5$	7.18×10^5	0.56276	$3{,}085.0 \pm 2{,}869.0$	9.44×10^3	0.30613
7.4	4.04×10^4	0.23701	$5{,}664.7 \pm 13{,}197.9$	9.47×10^4	0.16722	$28{,}623.3 \pm 95{,}611.4$	7.18×10^5	0.25089	$68{,}386.1 \pm 5.08\times10^5$	5.07×10^6	0.74086
46.9	2.47×10^5	0.07848	$22{,}522.2 \pm 4.64\times10^5$	1.46×10^7	0.64902	$18{,}981.2 \pm 90{,}062.0$	1.64×10^6	0.08626	$186{,}117.2 \pm 4.18\times10^6$	1.32×10^8	0.70702
19.8	3.33×10^6	0.09052	$28{,}023.4 \pm 1.57\times10^6$	1.56×10^8	0.55756	$31{,}355.1 \pm 5.80\times10^5$	5.35×10^7	0.17064	$161{,}899.9 \pm 4.91\times10^6$	4.16×10^8	0.25697
76.6	1.26×10^7	0.03399	$20{,}509.9 \pm 1.87\times10^6$	5.65×10^8	0.27533	$41{,}940.0 \pm 1.26\times10^6$	2.69×10^8	0.06418	$155{,}763.3 \pm 8.73\times10^6$	1.86×10^9	0.11951
$\times10^5$	2.18×10^8	0.05108	$15{,}669.2 \pm 8.35\times10^5$	5.65×10^8	0.03604	$53{,}072.5 \pm 3.01\times10^6$	1.27×10^9	0.02399	$182{,}519.6 \pm 2.32\times10^7$	1.66×10^{10}	0.09113
$\times10^5$	2.18×10^8	0.00538	$15{,}889.1 \pm 1.21\times10^6$	1.97×10^9	0.01238	$56{,}474.2 \pm 1.16\times10^7$	3.35×10^{10}	0.05933	$214{,}671.8 \pm 1.34\times10^8$	4.12×10^{11}	0.19173
0×10^5			16,384 $\pm 1.692\times10^9$			65,536 $\pm 1.501\times10^{15}$			262,144 $\pm 2.711\times10^{26}$		
= 58.7, max = 152			#visited: avg = 77.6, max = 202			#visited: avg = 99.2, max = 262			#visited: avg = 123.6, max = 296		

Table 8: IS by size/indegree, Klee-Minty cubes

Klee–Minty cubes												
	$n = 8$			$n = 10$			$n = 12$			$n = 14$		
m	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	
10^1	351.0 ± 409.4	1.43×10^3	0.40745	426.2 ± 297.3	1.01×10^3	0.23671	2,912.5 ± 3,257.5	9.80×10^3	0.33636	22,421.8 ± 55,586.2	1.80×10^5	
10^2	298.3 ± 361.5	2.80×10^3	0.09373	1,002.6 ± 2,417.0	2.08×10^4	0.20746	2,805.3 ± 3,774.8	2.08×10^4	0.07424	11,839.0 ± 23,835.6	1.80×10^5	
10^3	266.0 ± 341.4	4.58×10^3	0.01722	975.0 ± 1,852.5	2.49×10^4	0.02554	4,418.8 ± 20,658.1	4.78×10^5	0.10808	15,789.9 ± 1.50×10^5	4.68×10^6	
10^4	257.4 ± 345.6	6.25×10^3	0.00243	970.3 ± 2,375.3	8.89×10^4	0.00916	3,838.8 ± 17,371.0	1.21×10^6	0.03150	17,785.4 ± 3.10×10^5	2.37×10^7	
10^5	254.8 ± 368.4	1.64×10^4	0.00064	1,035.6 ± 8,173.8	2.13×10^6	0.02057	3,982.0 ± 27,084.6	4.81×10^6	0.01209	18,256.1 ± 6.65×10^5	2.01×10^8	
10^6	255.7 ± 389.0	3.87×10^4	0.00015	1,024.6 ± 4,759.7	2.13×10^6	0.00208	4,106.5 ± 50,004.1	3.63×10^7	0.00884	15,995.2 ± 2.97×10^5	2.01×10^8	
	256 #visited: avg = 74.0, max = 326 #generated: avg = 293.3			1,024 #visited: avg = 193.2, max = 1105 #generated: avg = 937.0			4,096 #visited: avg = 542.4, max = 4291 #generated: avg = 3092.3			16,384 #visited: avg = 1679.4, max = 15 #generated: avg = 10988.5		

Klee–Minty cubes												
$n = 12$			$n = 14$			$n = 16$			$n = 18$			
std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	
3,257.5	9.80×10^3	0.33636	22,421.8 ± 55,586.2	1.80×10^5	0.80175	11,926.7 ± 12,548.7	4.09×10^4	0.34294	65,721.0 ± 82,037.4	2.74×10^5	0.41725	
3,774.8	2.08×10^4	0.07424	11,839.0 ± 23,835.6	1.80×10^5	0.15184	28,482.4 ± 56,899.0	3.20×10^5	0.11236	53,239.6 ± 89,779.9	5.95×10^5	0.11184	
20,658.1	4.78×10^5	0.10808	15,789.9 ± 1.50×10^5	4.68×10^6	0.29638	44,626.2 ± 2.94×10^5	7.85×10^6	0.17598	116,193.8 ± 3.88×10^5	7.15×10^6	0.06151	
17,371.0	1.21×10^6	0.03150	17,785.4 ± 3.10×10^5	2.37×10^7	0.13301	48,505.9 ± 2.59×10^5	9.01×10^6	0.01858	415,364.8 ± 2.09×10^7	2.04×10^9	0.49150	
27,084.6	4.81×10^6	0.01209	18,256.1 ± 6.65×10^5	2.01×10^8	0.11022	54,430.6 ± 6.95×10^5	1.49×10^8	0.02739	267,091.4 ± 9.31×10^6	2.04×10^9	0.07644	
50,004.1	3.63×10^7	0.00884	15,995.2 ± 2.97×10^5	2.01×10^8	0.01258	65,950.8 ± 4.06×10^6	3.28×10^9	0.04968				
avg = 542.4, max = 4291 ed: avg = 3092.3			16,384 #visited: avg = 1679.4, max = 15203 #generated: avg = 10988.5			65,536 #visited: avg = 5778.3, max = 66363 #generated: avg = 42729.9			262,144 #visited: avg = 22382.7, max = 287258 #generated: avg = 185156.5			

Table 9: Bucket Sampling, Klee-Minty cubes

Klee–Minty cubes											
	$n = 8$			$n = 10$			$n = 12$			$n = 14$	
m	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.
10^1	262.2 ± 155.8	5.68×10^2	0.21673	698.9 ± 362.5	1.49×10^3	0.21311	1,747.4 ± 1,241.1	4.00×10^3	0.22862	4,492.1 ± 3,356.2	1.26×10^4
10^2	265.7 ± 187.3	1.30×10^3	0.04902	1,010.6 ± 1,053.9	6.45×10^3	0.06381	3,533.0 ± 6,814.3	5.70×10^4	0.16136	9,006.2 ± 10,698.2	5.26×10^4
10^3	256.2 ± 178.2	1.70×10^3	0.00664	1,006.2 ± 1,586.2	2.99×10^4	0.02973	3,861.8 ± 11,011.9	2.55×10^5	0.06613	16,390.4 ± 75,235.8	1.71×10^6
10^4	256.5 ± 172.8	2.04×10^3	0.00079	1,023.7 ± 1,559.8	5.64×10^4	0.00551	4,172.8 ± 12,451.9	5.04×10^5	0.01208	17,259.7 ± 1.45×10^5	1.26×10^7
10^5	255.7 ± 176.4	3.35×10^3	0.00013	1,019.5 ± 1,446.9	7.24×10^4	0.00071	4,107.1 ± 11,862.7	8.52×10^5	0.00207	16,395.5 ± 1.04×10^5	1.26×10^7
10^6	255.8 ± 178.0	7.22×10^3	0.00003	1,022.6 ± 1,490.0	2.40×10^5	0.00023	4,104.1 ± 13,739.7	3.46×10^6	0.00084	16,410.4 ± 1.25×10^5	4.84×10^7
	256 #visited: avg = 508.7, max = 919 #generated: avg = 1450.7 #saved: avg = 358.6 = 24.7%			1,024 #visited: avg = 734.0, max = 1381 #generated: avg = 2559.3 #saved: avg = 408.3 = 16.0%			4,096 #visited: avg = 971.0, max = 1880 #generated: avg = 4012.0 #saved: avg = 433.3 = 10.8%			16,384 #visited: avg = 1221.2, max = 23 #generated: avg = 5836.0 #saved: avg = 448.9 = 7.7%	

Klee–Minty cubes											
$n = 10$			$n = 12$			$n = 14$			$n = 16$		
± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$	mean ± std. dev.	max.	$\hat{\chi}$
9 ± 362.5	1.49×10^3	0.21311	1,747.4 ± 1,241.1	4.00×10^3	0.22862	4,492.1 ± 3,356.2	1.26×10^4	0.27975	12,715.0 ± 20,826.3	6.84×10^4	0.53757
6 ± 1,053.9	6.45×10^3	0.06381	3,533.0 ± 6,814.3	5.70×10^4	0.16136	9,006.2 ± 10,698.2	5.26×10^4	0.05840	46,147.7 ± 1.15×10^5	8.48×10^5	0.18378
2 ± 1,586.2	2.99×10^4	0.02973	3,861.8 ± 11,011.9	2.55×10^5	0.06613	16,390.4 ± 75,235.8	1.71×10^6	0.10456	48,799.9 ± 1.83×10^5	3.48×10^6	0.07122
7 ± 1,559.8	5.64×10^4	0.00551	4,172.8 ± 12,451.9	5.04×10^5	0.01208	17,259.7 ± 1.45×10^5	1.26×10^7	0.07292	57,899.0 ± 2.90×10^5	9.64×10^6	0.01665
5 ± 1,446.9	7.24×10^4	0.00071	4,107.1 ± 11,862.7	8.52×10^5	0.00207	16,395.5 ± 1.04×10^5	1.26×10^7	0.00768	63,761.1 ± 7.61×10^5	1.30×10^8	0.02037
6 ± 1,490.0	2.40×10^5	0.00023	4,104.1 ± 13,739.7	3.46×10^6	0.00084	16,410.4 ± 1.25×10^5	4.84×10^7	0.00295	63,744.1 ± 1.13×10^6	5.44×10^8	0.00854
ed: avg = 734.0, max = 1381 rated: avg = 2559.3 d: avg = 408.3 = 16.0%			4,096 #visited: avg = 971.0, max = 1880 #generated: avg = 4012.0 #saved: avg = 433.3 = 10.8%			16,384 #visited: avg = 1221.2, max = 2340 #generated: avg = 5836.0 #saved: avg = 448.9 = 7.7%			65,536 #visited: avg = 1482.7, max = 2960 #generated: avg = 8046.2 #saved: avg = 460.4 = 5.7%		

Table 10: Crowd Sampling, budget $B = 10$, Klee-Minty cubes